

Chapter 5: Elasticity and Its Application

The Price Elasticity of Demand

The **price elasticity of demand** measures how much the **quantity demanded** of a good responds to a **change in its price**. Informally speaking, it measures **how price-sensitive buyers are**.

Factors that Determine the Price Elasticity of Demand

- **Availability of close substitutes**: Goods with close substitutes have more elastic demand, because buyers can easily switch. (E.g. if the price of one brand of instant noodles rises, we can buy another brand)
- **Necessities vs luxuries**: Necessities tend to have less elastic demand than luxuries (E.g. even if the cost of a doctor's visit rises, we still have to go just as often)
- **Definition of the market**: Demand is usually more elastic to a **narrowly defined good**
- **Time horizon**: Demand is usually more elastic over longer periods (E.g. when gasoline gets more expensive, drivers can cut back only a little in the first few months, over years, they can move closer to work)

The Variety of Demand Curves

Key Idea

Economists classify demand curves according to their **elasticity**.

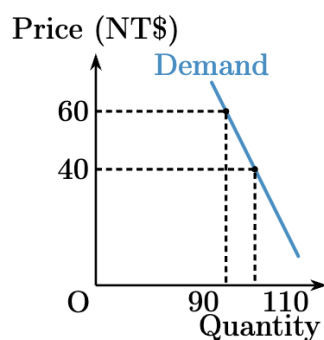
- Demand is **elastic** when the **elasticity > 1** , so quantity moves proportionally **more** than price
- Demand is **inelastic** when **elasticity < 1** , so quantity moves proportionally **less** than price
- Demand is **unit elastic** when **elasticity $= 1$** , so quantity and price change by **equal percentages** in absolute value

Remark: With the same axis units, the **flatter** the demand curve, the **greater the price elasticity of demand**. The **steeper** it is, the **smaller the price elasticity of demand**.

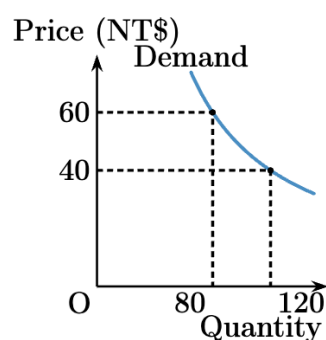
(a) Perfectly inelastic:
elasticity 0



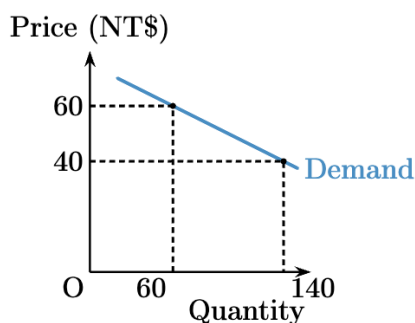
(b) Inelastic: elasticity
below 1



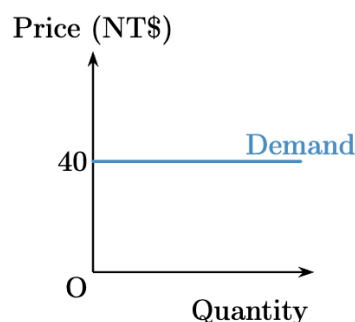
(c) Unit elastic:
elasticity 1



(d) Elastic: elasticity above 1



(e) Perfectly elastic: elasticity infinite



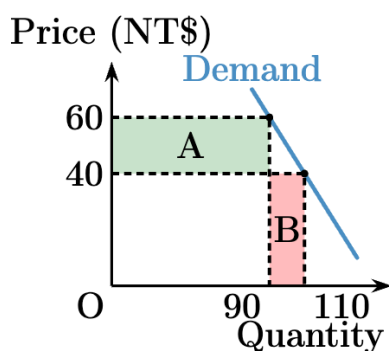
Total Revenue

Key Idea

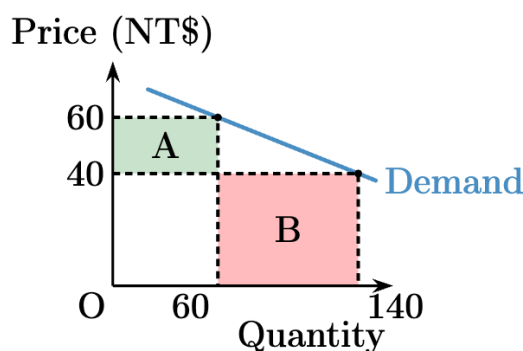
Total revenue is the amount sellers receive from sales: the price received per unit times the quantity sold $P \times Q$. Without a *tax wedge*, this also equals buyers' total spending. Graphically, it's the **area of the box** under the demand curve.

Here, we can see how total revenue (i.e. shaded area) changes when price changes, according to if demand is inelastic or elastic:

(a) Inelastic demand



(b) Elastic demand



The Elasticity–Revenue Rule

Along a fixed demand curve, **elasticity** determines whether the **price effect** or the **quantity effect** on total revenue, $P \times Q$, is **stronger**.

Demand	Price rises	Price falls
Inelastic ($E < 1$)	Revenue rises	Revenue falls
Unit elastic ($E = 1$)	Revenue unchanged	Revenue unchanged
Elastic ($E > 1$)	Revenue falls	Revenue rises

Summary

With inelastic demand, price and revenue move in the same direction. With elastic demand, they move in opposite directions.

Notice that when asked about if the total revenue of a good would increase or fall, the price change alone does not determine the answer—make sure to talk about the elasticity.

Elasticity along a Linear Demand Curve

A straight line has constant slope, but **not constant elasticity**, since elasticity is a ratio of percentage changes.

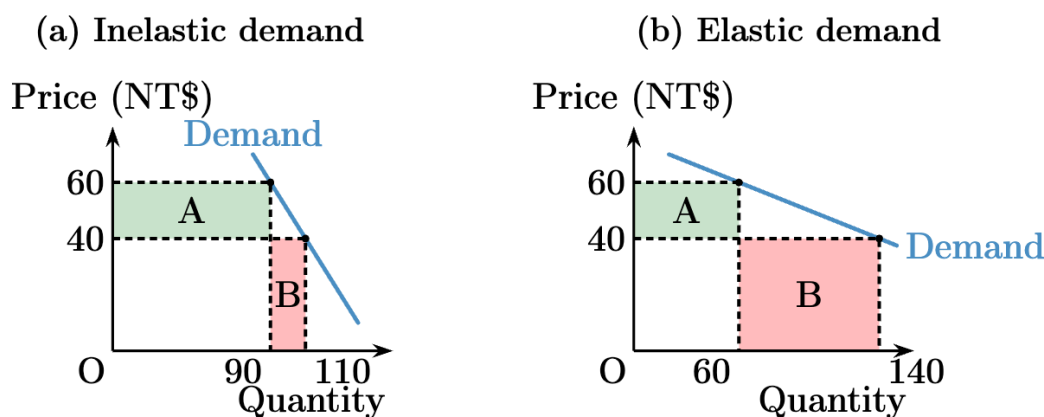
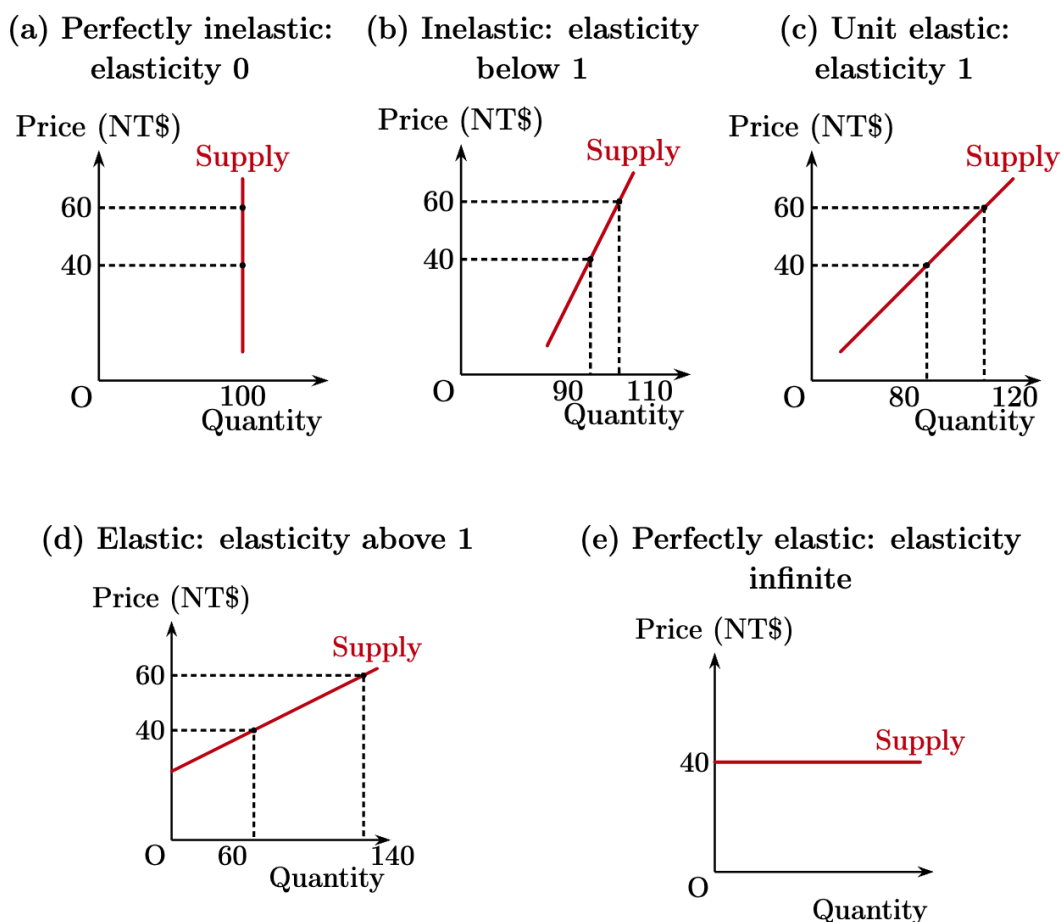


Figure 14: Elasticity along a Linear Demand Curve

The Price Elasticity of Supply

The **price elasticity of supply** measures how much the **quantity supplied** of a good **responds to a change in its price**. Informally, it is **how flexible** sellers are in changing the amount they produce.



Applications

Consider a hypothetical competitive rice market with no imports, exports, or government price support. Suppose a new rice variety lowers production costs without changing the rice's quality. Farmers can supply more rice at every price, so **supply shifts right**. The **equilibrium price falls** and **quantity rises**.

Farmers sell more rice but receive less per kilogram. However, rice may have low demand elasticity, so the **total revenue may decrease**. Although adopting the new variety is individually profitable at the original market price, when all farmers adopt it, the resulting increase in market supply drives the equilibrium price down. The effect on profit still depends on the reduction in production costs.

Key Idea

What is optimal for each individual may not be optimal for the group as a whole.